# Digital Electronics I

[EE1-02]

**Question 1**

1. a) (i)

$$\overline{\overline{A}+\overline{(A\overline{B})}\,\overline{C}}$$
$$= A\overline{\overline{(A\overline{B})}\,\overline{C}}$$
$$= A(A\overline{B}+C)$$
$$= A\overline{B}+AC$$

[4]

ii)

$$ABC + B(A \oplus B)$$
$$= ABC + B(\overline{A}B + A\overline{B})$$
$$= ABC + \overline{A}B$$
$$= B(AC + \overline{A})$$
$$= B(C + \overline{A})$$
$$= BC + \overline{A}B$$

[4]

b)

$$f = \overline{A}\overline{B}\overline{(C \oplus D)} + A\overline{B}\overline{C}\overline{D} + ACD + ABD$$
$$= \overline{A}\overline{B}CD + \overline{A}\overline{B}\overline{C}\,\overline{D} + A\overline{B}\overline{C}\overline{D} + ACD + ABD$$

| f | CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|---|
| AB | | | | | |
| 00 | | 1 | 0 | 1 | 0 |
| 01 | | 0 | 0 | 0 | 0 |
| 11 | | 0 | 1 | 1 | 0 |
| 10 | | 1 | 0 | 1 | 0 |

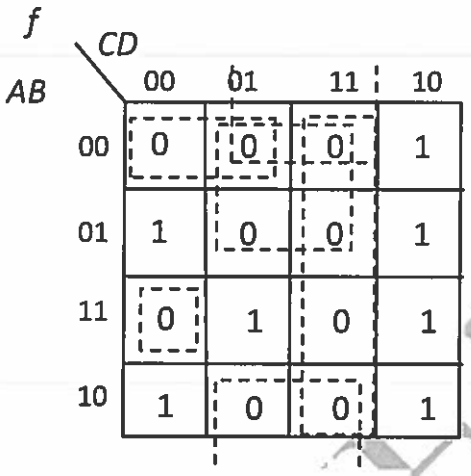
$$f = \overline{B}\overline{C}\overline{D} + ABD + \overline{B}CD$$

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Here, 1 mark for drawing the Karnaugh map, 1 for filling it out correctly, 1 for the correct grouping, and 1 for the final expression.

[4]

c)




$$f = \left(A + \overline{D}\right)\left(B + \overline{D}\right)\left(A + B + C\right)\left(\overline{A} + \overline{B} + C + D\right)\left(B + \overline{C} + \overline{D}\right)$$

Here, 1 mark for drawing the Karnaugh map, 1 for filling it out correctly, 1 for the correct grouping, and 1 for the final expression.

[4]

d)

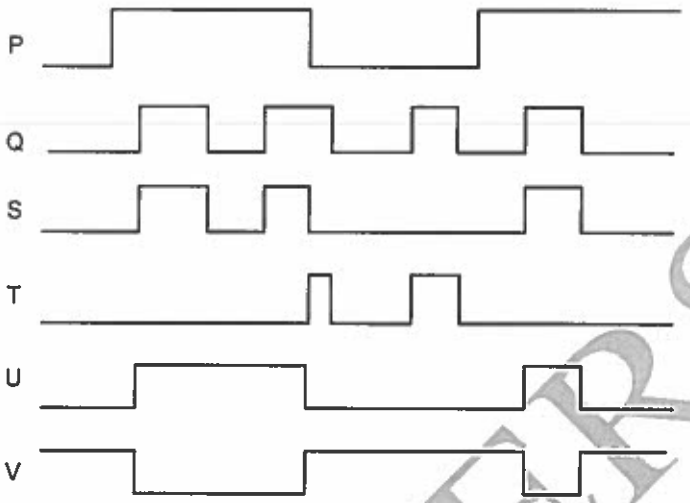
| Decimal | Hexadecimal | Binary | BCD |
|---|---|---|---|
| 5153 | 1421 | | |
| | | 0011 0111 1001 1000 | (1) 0100 0010 0011 0010 |
| | 1CF | 0000 0001 1100 1111 | |
| -1101 | | 1111 1011 1011 0011 | |

Give 2 marks per answer.

[8]

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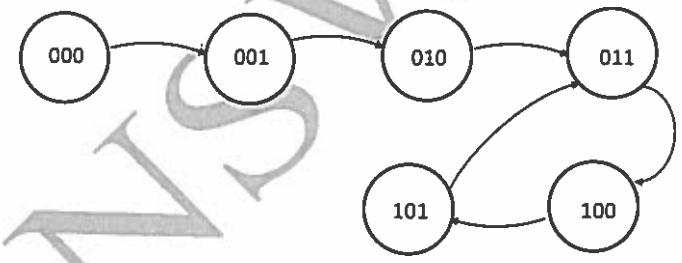
e)




Marks: 2 marks per waveform

[8]

f)




Give 2 marks correct states and 2 marks for correct interconnections.

[4]

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g)

| A | B | C | Y | Z |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Give 0.5 marks per row of the table.

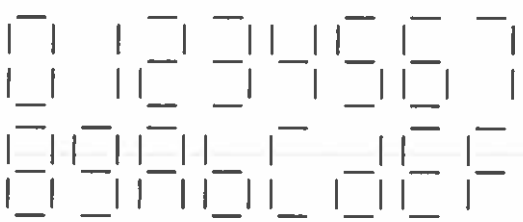
[4]

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**Question 2**

2.

(a) (i) The hexadecimal pattern is as follows:



[4]

(ii) We first draw the K-map for the top horizontal segment, with the hexadecimal number corresponding to the variables X[3:0]:

| X3, X2 \ X1, X0 | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 1 | 1 |
| 01 | 0 | 1 | 1 | 1 |
| 11 | 1 | 0 | 1 | 1 |
| 10 | 1 | 1 | 0 | 1 |

This gives the Boolean expression:

$$f = \overline{X0}X3 + X1X2 + X1\overline{X3} + \overline{X0}\,\overline{X2} + \overline{X1}X2X3 + X0X2\overline{X3}$$

The expression may be converted directly to NAND gates only by applying De Morgan's theorem:

$$f = \overline{\overline{X0}X3 + X1X2 + X1\overline{X3} + \overline{X0}\,\overline{X2} + \overline{X1}X2X3 + X0X2\overline{X3}}$$
$$= \overline{\left(\overline{X0}X3\right)\left(X1X2\right)\left(X1\overline{X3}\right)\left(\overline{X0}\,\overline{X2}\right)\left(\overline{X1}X2X3\right)\left(X0X2\overline{X3}\right)}$$

The circuit then uses 4 two-input NAND gates, 2 three-input NAND gates, 1 six-input NAND gate, and 7 inverters.

Marks: 1 marks for K-map, 1 marks for initial expression from this, 2 marks for NAND expression.

[4]

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(iii)

| | | X1, X0 | | | |
|---|---|---|---|---|---|
| | $f$ | 00 | 01 | 11 | 10 |
| X3, X2 | 00 | 1 | 0 | 1 | 1 |
| | 01 | 0 | X | 1 | 1 |
| | 11 | 1 | 0 | 1 | 1 |
| | 10 | 1 | X | 0 | 1 |

For a NOR implementation, we can solve in POS form:

$$\overline{f} = \overline{X1 + \overline{X0}} + \overline{X3 + \overline{X2} + X1} + \overline{\overline{X3} + X2 + \overline{X0}}$$

$$\Rightarrow f = \overline{\overline{\left(X1 + \overline{X0}\right)} + \overline{\left(X3 + \overline{X2} + X1\right)} + \overline{\left(\overline{X3} + X2 + \overline{X0}\right)}}$$

Marks: 1 marks for K-map, 1 mark for the grouping and/or an initial expression, and 2 marks for the final NOR expression.

[4]

b) To determine if A = B requires a bit-by-bit comparison using an XNOR gate will determine equality. To determine A > B again requires a bit-by-bit comparison. Starting from the most significant bit, if $A_3 > B_3$ then A > B. However, if this is not true, and $A_3 = B_3$, then we need to check the next set of bits and so on. This gives the following Boolean expressions for H, E and L:

$$H = A_3\overline{B}_3 + \left(\overline{A_3 \oplus B_3}\right)A_2\overline{B}_2 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)A_1\overline{B}_1 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\left(\overline{A_1 \oplus B_1}\right)A_0\overline{B}_0$$

$$E = \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\left(\overline{A_1 \oplus B_1}\right)\left(\overline{A_0 \oplus B_0}\right)$$

$$L = \overline{A}_3B_3 + \left(\overline{A_3 \oplus B_3}\right)\overline{A}_2B_2 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\overline{A}_1B_1 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\left(\overline{A_1 \oplus B_1}\right)\overline{A}_0B_0$$

Alternatively, given H and E as above, we have: $L = \overline{H}.\overline{E}$

Give 2 marks for identifying logic, and 2 marks each for the three Boolean expressions.

[8]

c) The 8 bit numbers A[7:0] and B[7:0] can each be broken up into two halves containing the upper four and lower four bits. We can then use COMP modules to compare A[7:4] with B[7:4], and A[3:0] with B[3:0].

[2]

Additional combinational logic is necessary to allow the H, E and L outputs of each COMP module to be compared and obtain the comparison between A[7:0] and B[7:0]. The following logic will produce H, E and L:

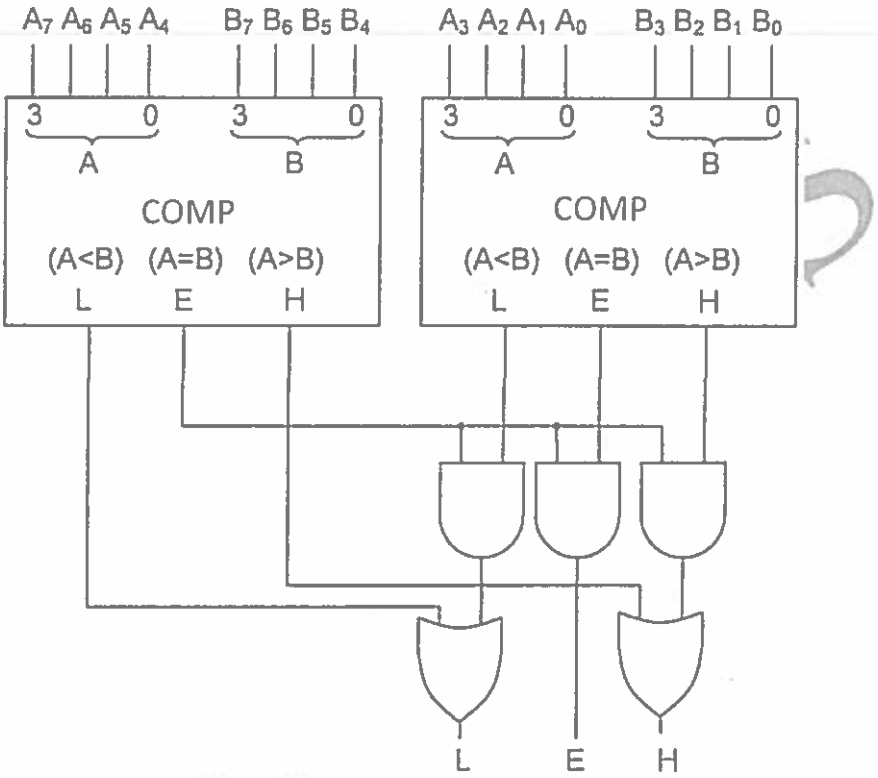
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H = 1 if A[7:4] > B[7:4], OR: A[7:4] = B[7:4] AND A[3:0] > B[3:0].
L = 1 if A[7:4] < B[7:4], OR: A[7:4] = B[7:4] AND A[3:0] < B[3:0].

E = 1 if A[7:4] = B[7:4] AND A[3:0] = B[3:0].

[4]

This gives the following circuit diagram:




[4]

**Question 3**

3. a) With A, B as the present state of the FSM, $A^+$, $B^+$ as the next state, X input and Y output, we have the following state transition table:

| A | B | X | $A^+$ | $B^+$ | $Y^+$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 |

Give 2 marks for basic table format and 0.5 marks for each correct row of the table.

[6]

b) For an FSM implemented using D-flip-flops, the next state of the flip-flop is simply the present input to the flip-flop. This gives the following Karnaugh maps and Boolean equations:

For $A^+$:

| X \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 |

$$\Rightarrow \quad A^+ = \overline{X}B + X\overline{AB}$$

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For $B^+$:

| X \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 |

$$\Rightarrow \quad B^+ = X\overline{B} + \overline{X}AB$$

For Y

| X \ AB | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 |

$$\Rightarrow \quad Y = X\overline{B}$$

Give 1 mark for each K-map and 1 mark for each Boolean expression.

[6]

c) Using the result of (b) it can be seen that 2 OR gates and 4 AND gates are necessary. Note that the Y term does not need an additional AND gate as it is available within the expression for $B^+$.

[4]

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d) The state transition table may be redrawn with one-hot encoding as follows:

| S3S2S1S0 | X | $S3^+$ $S2^+$ $S1^+$ $S0^+$ | $Y^+$ |
|---|---|---|---|
| 0001 | 0 | 0001 | 0 |
| 0001 | 1 | 0010 | 1 |
| 0010 | 0 | 0100 | 0 |
| 0010 | 0 | 0001 | 0 |
| 0100 | 0 | 0001 | 0 |
| 0100 | 1 | 1000 | 1 |
| 1000 | 0 | 1000 | 0 |
| 1000 | 1 | 0001 | 0 |

Give 4 marks for understanding this and for drawing the table.

By inspection of the table:

$$S_3^+ = S_2X + S_3\overline{X}$$

$$S_2^+ = S_1\overline{X}$$

$$S_1^+ = S_0X$$

$$S_0^+ = S_0\overline{X} + S_1X + S_2\overline{X} + S_3X$$

$$Y = S_0X + S_2X$$

Give 2 marks for each expression.

[14]

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